

Arsh Shivayach

Software Engineer

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Profile

Software Engineer with 5+ years of experience developing high-performance C++ and Qt desktop applications, including performance-critical and crash-sensitive systems. Skilled in debugging, profiling, crash analysis, performance tuning, and memory safety. Experienced in AI-assisted development, using a specification-first workflow to define executable requirements, supervise AI agents for implementation and refactoring, and validate correctness through structured review and testing. Strong background collaborating across architecture, QA, and engineering to ensure reliable delivery and maintainable codebases. Hands-on with Windsurf (Claude Sonnet 4.5), Agent Copilot workflows, GitHub Copilot, ChatGPT, and Gemini for accelerated software development and automation.

WORK EXPERIENCE

2023 – Present

Graebert GmbH

Software Engineer

- Contributed to the development and enhancement of **performance-sensitive C++/Qt CAD software**, including geometry processing, UI behavior, and platform workflows.
- Shifted engineering approach from manual implementation to a **specification-first, AI-supervised delivery model**, improving consistency and reducing development effort.
- Utilized **Windsurf (Claude Sonnet 4.5), GitHub Copilot, ChatGPT, Codeium, and Gemini** for coding assistance, debugging, refactoring, automation, and feature implementation.
- Authored **executable specifications, technical requirements, and architectural guidelines** used to drive agent-based development workflows.
- Performed debugging, crash analysis, regression prevention, and performance tuning to improve software reliability and responsiveness.
- Collaborated with QA teams to validate expected behavior and ensure release-ready builds with controlled iteration and structured review.
- Supported organization-wide onboarding and adoption of **AI-accelerated workflows**, including documentation, workflow refinement, and best-practice alignment.

2022 – 2023

Agiliad Technologies

Senior Software Engineer

- **Designed and implemented C++-based imaging workflows**, integrating Fiji ImageJ and pugiXML to support automated processing, data extraction, and configurable imaging pipelines.
- **Improved legacy modules by developing automation utilities and modernizing architecture**, resulting in better maintainability, reduced manual effort, and improved scalability for future feature additions.
- **Collaborated across engineering and QA teams to perform design reviews, debugging, and release validation**, ensuring regression-safe delivery and stable production rollout.

Amdocs India Development Center LLP

Software Engineering Associate

- **Delivered enhancements, features, and defect fixes in C++ and SQL-based enterprise applications**, working from high-level requirements and standardized testing criteria.
- **Participated in the full SDLC**, including requirement clarification, implementation, code reviews, CI workflows, and coordinated release pipelines.
- **Supported production readiness and post-release stability**, including debugging, patch creation, and incremental controlled rollouts in large-scale enterprise environments.

EDUCATION

2020 – 2020

PG Diploma in Advanced Computing

CDAC - ACTS

2015 – 2019

Bachelor of Technology in Electronics and Communication Engineering

Chandra Shekhar Azad University, Kanpur

SKILLS

Programming & Engineering

C++, Qt, STL, OOP, Design Patterns, SQL, Shell Scripting

Debugging & Delivery

Crash analysis, performance tuning, memory safety, SPEC Development

Development Approach

Specification-first mindset, test-driven delivery, regression-safe integration

Version Control & Workflow

Git, Perforce, CI/CD, code reviews, Jira

AI-Assisted Development

Windsurf (Claude Sonnet 4.5), GitHub Copilot, ChatGPT, Gemini

Platforms

Linux, Windows

The Coding Evolution Story

- Began with traditional development focused on **debugging minor issues**.
- Progressed to **isolated defect fixing with AI guidance**.
- Leveraged AI to create **partial feature scaffolding** and foundational structures.
- Advanced to **full feature implementation driven by clear technical specifications**.
- Reached a stage where **coordinated AI agents deliver end-to-end features** with human architectural oversight.
- Shifted engineering focus from manual coding to **system-level reasoning, precise specifications, clarity, and correctness**, while AI handles **implementation, refactoring, testing, and iteration**.

Interests

- Chess
- Rubik's Cube
- Sudoku
- Teaching / Training
- Problem Solving